Perry–Beurling Spectral Sieve (PBSS): Status Note on Lemmas, Open Plateau, and Theorem-A Scaffolding

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Independent research note — not a proof of RH
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Abstract

We summarize the reconstructed **Perry–Beurling Spectral Sieve (PBSS)**: a projection diagnostic for density residuals on log-windows. Model theorems A_0/B_0 are proved as lemmas M1–M5; admissible weights preserve model decay (M6). A grand campaign and open-plateau interventions show the arithmetic residual soft-plateaus at $R_d \approx 0.15$–0.19 through $x \sim 5 \times 10^{10}$ and does *not* exhibit A_0-style collapse under peel, detrend, or dense multi-T. Theorem-A scaffolding (weight class W_α, truncation remainder, tail majorants) is shipped; full A/B and RH remain open. This note is a status brief, not a claim of RH.

1 Setup

Fix a logarithmic window length $T > 0$ and set $x = e^{uT}$ for $u \in [0, 1]$. Let $\{\varphi_k\}_{k\geq 0}$ be the shifted Legendre orthonormal system on $L^2([0, 1], du)$:

$$\varphi_k(u) = \sqrt{2k+1}\, L_k(2u-1), \qquad V_d = \operatorname{span}\{\varphi_0, \ldots, \varphi_d\}, \quad P_d = \text{proj. onto } V_d.$$

For a residual $q \in L^2([0, 1])$ define the energy ratio

$$R_d(q) = \frac{\|P_d q\|_{L^2}^2}{\|q\|_{L^2}^2} = \frac{\sum_{k=0}^{d} |\langle q, \varphi_k\rangle|^2}{\|q\|^2} \in [0, 1], \qquad S_d(q; T) = T^{2(d+1)} R_d(q).$$

Low R_d: high-frequency content (RH-like for pure critical-line modes). High R_d: persistent low-degree mass (defect).

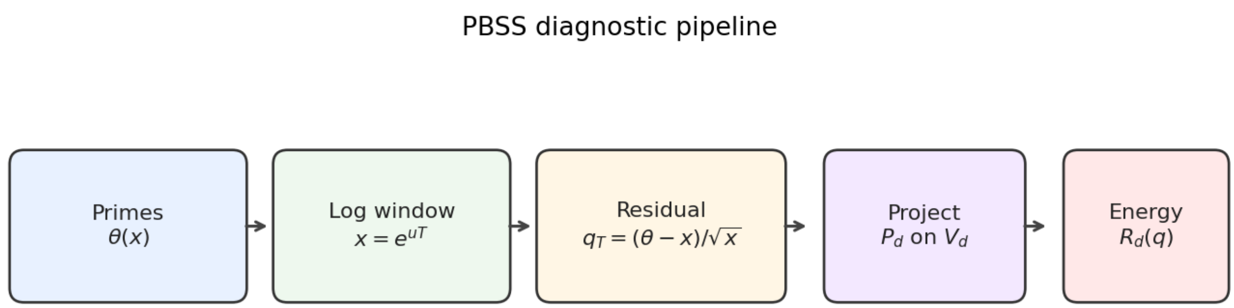

Figure 1: Diagnostic pipeline: Chebyshev residual on a log-window, projected onto low-degree Legendre modes; report R_d.

Model residuals. Critical-line (CL) mode $q_T^{\text{cl}}(u) = \sin(tTu)$; off-critical $e^{T(\sigma-1/2)u}\sin(tTu)$; orthogonal defect $q = \sqrt{1-\varepsilon^2}\, f + \varepsilon\,\varphi_j$ with $f \perp V_d$, $\|f\|_2 = 1$.

Arithmetic residual (shipped probe). With $\theta(x) = \sum_{p\le x} \log p$,

$$q_T(u) = \text{detrend}\Big(\frac{\theta(e^{uT}) - e^{uT}}{\sqrt{e^{uT}}}\Big) \quad \text{(default detrend: degree one).}$$

2 Proved diagnostic lemmas ($\mathbf{A}_0$/$\mathbf{B}_0$/M5/M6)

Lemma	Statement
M1	$R_d(\varphi_m) = 1_{m\le d}$
M2	Orthogonal defect $\Rightarrow R_d(q) = \varepsilon^2$
M3	CL pure mode $R_d(\sin(tTu)) = O_d(T^{-2})$ **($\mathbf{A}_0$)**
M4	Fixed $\varepsilon > 0 \Rightarrow R_d \not\to 0$ (blocks vanishing)
M5	Finite CL sum $R_d = O_d(T^{-2})$ (finite-mode A_0)
M6	Admissible weight w: $R_d(wq) = O_d(T^{-2})$ on model CL/EF

M1–M6 are continuous L^2 arguments (`src/pbss/lemmas.py`, `docs/PROOFS_LEMMAS.md`) with discrete verification in `tests/`.

- **$\mathbf{A}_0$ / finite-mode $\mathbf{A}_0$ (proved):** pure and finite CL residuals decay.
- **Weighted model $\mathbf{A}_0$ (proved, M6):** Tukey/Hanning weights preserve decay.
- **$\mathbf{B}_0$ (proved):** fixed low-degree defect blocks vanishing.

None of this proves arithmetic Theorem A or RH.

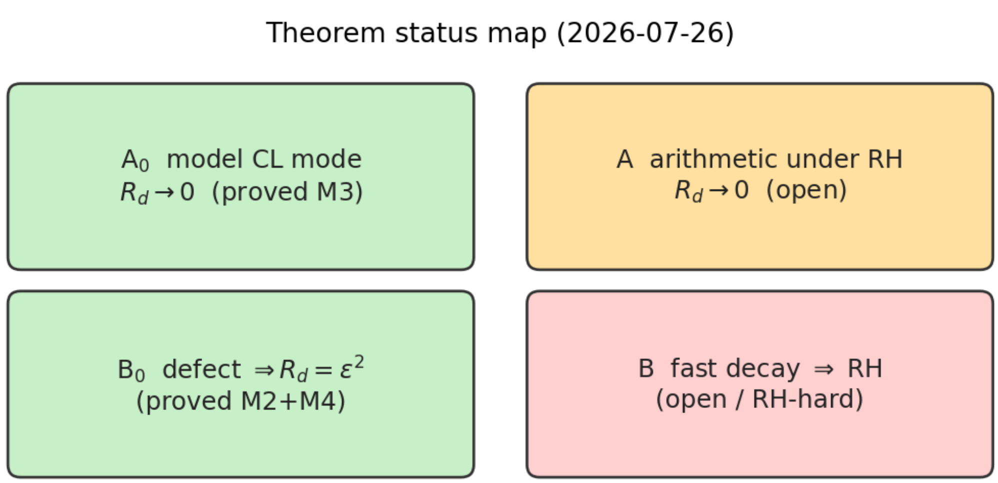

Figure 2: Status map. Green: proved model results. Amber/red: full A/B open.

3 Theorems A and B (open beyond models)

Full Theorem A (conditional on RH; open). Under RH, for the arithmetic residual q_T (or an equivalent explicit-formula residual), $R_d(q_T) \to 0$ as $T \to \infty$. Outline: RH $\Rightarrow$ superposition of CL modes; each contributes $O(T^{-2})$ by M3; controlling the sum over zeros is analytic number theory not completed here.

Full Theorem B (open; RH-hard). If $R_d(q_T)$ decays sufficiently fast, then ζ has no zero off the critical line. The implication is essentially as hard as RH; not proved in this repository.

4 Grand campaign numerics

Runner: `experiments/run_grand_campaign.py`. Scale: $x_{\max} = 10^{10}$ (455 052 511 primes, segmented sieve + on-disk checkpoint); 14 windows in T; ablations $d \in \{2, 4, 6, 8\}$, detrend $\in \{\text{none}, \text{deg0}, \text{deg1}\}$; controls CL / off-critical σ=0.75 / defect ε=0.5; **20 000** MC spectral-defect trials per T (280 000 total); wall $\sim$ 1 hour multi-core; resume-capable state JSON.

Table 1: Focus cut ($d = 4$, deg1) and controls at large T.

T	$x_{\max}$	R_4 arith	R_4 CL	R_4 defect	MC mean
10	2×10^4	0.144	$\sim6\times10^{-3}$	0.250	0.793
16	9×10^6	0.193	$\sim2\times10^{-3}$	0.250	0.793
23	1×10^{10}	**0.155**	$\sim5\times10^{-4}$	0.250	0.795

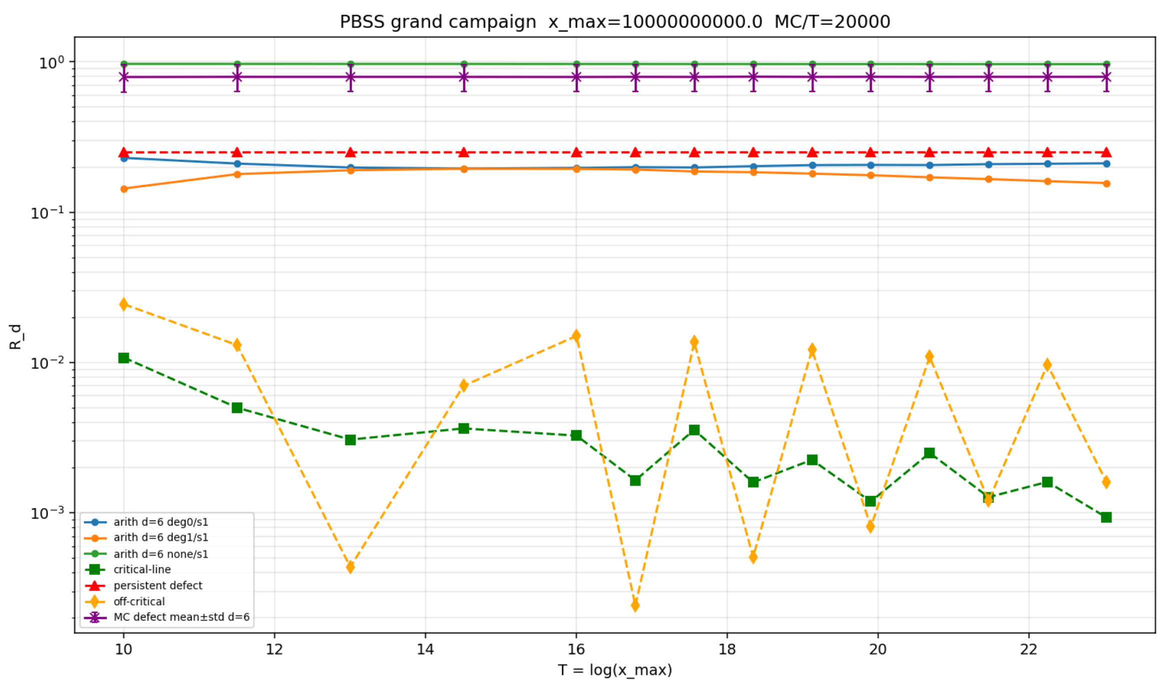

Figure 3: Grand campaign: arithmetic ablations, CL/defect/off-critical controls, MC defect mean $\pm$ std. Arithmetic tracks stay $O(10^{-1})$; CL decays; defects stay high.

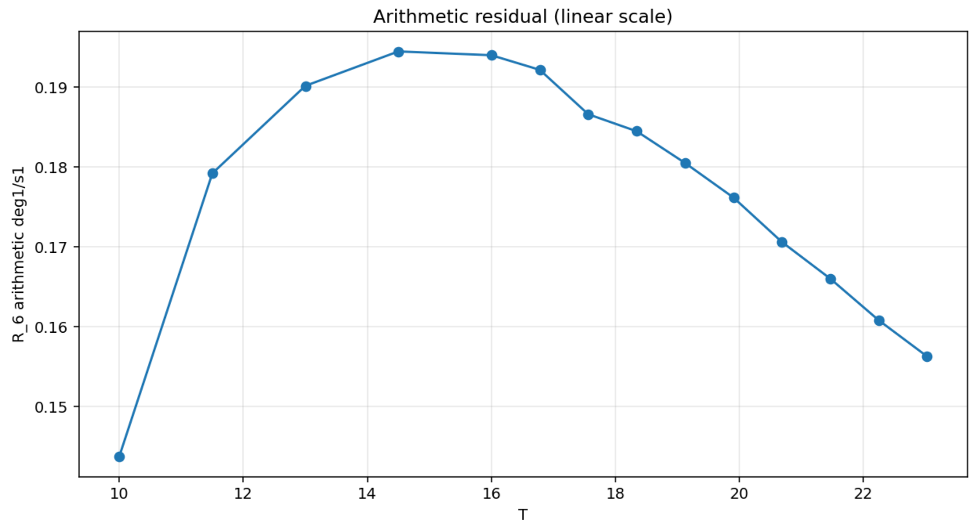

Figure 4: Arithmetic focus on a linear R_d scale: soft plateau ~ 0.15–0.19 through $T \approx 23$ ($x_{\max} = 10^{10}$).

Reading.

1. The diagnostic separates pure CL modes from planted defects as $\mathrm{A}_0/\mathrm{B}_0$ predict.
2. Arithmetic $q_T = (\theta - x)/\sqrt{x}$ (detrended) does **not** collapse like A_0 through ten billion; it soft-plateaus.
3. This does **not** prove or disprove RH. The current arithmetic probe is not yet the object for which full Theorem A is numerically visible.

5 Open-plateau interventions (to $x \sim 5 \times 10^{10}$)

Runner: `experiments/run_open_plateau.py`; synthesis: `docs/RESEARCH_PLATEAU.md`.

Seven intervention classes (peel, whiten, measure, basis, scale, MC, Beurling). Deep axes: 50M MC trials; dense multi-T residual rebuilds on the full 5×10^{10} table; Beurling battery ≥ 500 systems.

Headline numbers (deg1 arithmetic). Focus $R_4 \sim 0.15$–0.19 through $T \approx \log(5 \times 10^{10})$. Peeling more model zeros does **not** collapse R_d. Detrend is required; edge taper lowers R_d (endpoint mass). Dense multi-T shows no A_0 decay.

6 Theorem-A scaffolding

- **Weight class W_α:** Tukey/Hanning; endpoint estimator E_{end} (`pbss.weights`). Model decay under weights: Lemma M6.
- **Truncation remainder:** peel $q - \alpha q_T^{(N)}$; M5 majorant on finite blocks; scaffolding infinite-tail majorant (`pbss.remainder`, `docs/INFINITE_TAIL_REMAINDER.md`).
- **Arithmetic weights campaign:** multi-T raw vs weighted R_d and E_{end} on available primes (`results/arithmetic_w`

Writeup: `docs/THEOREM_A_SCAFFOLD.md`. Full arithmetic A still open (infinite zero sum + true ψ/θ remainder + weight-class theorem for arithmetic q_T).

7 Done vs open

Done	**Open**
Lemmas M1–M6; A_0/B_0/weighted model A_0	Full Theorem A (arithmetic under RH)
Sieve + checkpoint to 5×10^{10}	Full Theorem B
Open-plateau catalog + deep MC/scale/Beurling	Sharp zero-density tail under ζ
Weight class + truncation remainder scaffold	Arithmetic EF remainder with constants
Tests green; status note refreshed	Legacy $P \approx 3.92$ normalization

8 Explicit non-claim

> *This repository and this note do not contain an unconditional proof of the Riemann Hypothesis. Proved results are about the diagnostic (A_0/B_0/M5/M6). Full A/B and RH remain open.*

Artifacts: `docs/STATUS.md`, `THEOREMS_AB.md`, `PROOFS_LEMMAS.md`, `RESEARCH_PLATEAU.md`, `THEOREM_A_SCAFFOLD.md`, `INFINITE_TAIL_REMAINDER.md`, `results/grand_campaign/`, `results/open_plateau/`, `results/arithmetic_weights`
Independent research; MIT licensed. Not a proof of RH.